Nilesh Raj

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Summary

AI/ML Engineer with hands-on experience in NLP, fine-tuning transformer models, and agentic AI systems. Built and deployed BERT text classifiers, LangGraph state machine RAG pipelines, and CrewAI multi-agent platforms in production. Experienced with PyTorch, Scikit-learn, Hugging Face, LangChain, vector databases, and FastAPI microservices. NVIDIA-certified in LLM application development.

Education

United University Prayagraj — Prayagraj, UP
BCA with IBM Specialization (AI & Data Science) — *Jun. 2022 – May 2025*

Projects

OMNICREW AI | *Python, CrewAI, LangChain, FastAPI, Redis, Docker* — 2026

- Orchestrated a multi-agent pipeline using CrewAI with a 3-tier LLM fallback router (DeepSeek → Mistral AI → Gemini), achieving 99.9% task completion across real-time web extraction tasks.
- Engineered Redis-backed MD5 caching for LLM responses, eliminating duplicate inference calls and reducing latency from 2.5s to under 50ms with zero additional token cost on cache hits.
- Implemented semantic deduplication using MD5 hashing and INSERT IGNORE logic, reducing redundant database writes by an estimated 35%.
- Deployed App | Code

LumanGuide | *Python, LangGraph, FAISS, FastAPI, Streamlit, LangSmith* — 2026

- Architected an Agentic RAG system on a LangGraph state machine with dynamic document grading and query routing, cutting retrieval latency by 30% over a naive RAG baseline.
- Built a FAISS vector store with semantic chunking and embedding models for sub-second document retrieval and migrated user auth from SQLite to MongoDB Atlas, eliminating 100% of data-wipe incidents.
- Hardened with prompt injection guardrails, JWT RBAC, and automated regex secret scanning, achieving 100% credential redaction before vector embedding.
- Integrated LangSmith for LLMOps observability, tracing retrieval accuracy, token usage, and pipeline latency across all queries. Live App | Code

Experience

AI/ML Trainee (Internship) — Feb. 2026 – May 2026
Technosavvys Education Technology — *Remote*

- Fine-tuned a BERT text classifier on 500+ student reviews using Hugging Face, achieving 88% accuracy (a 15% lift over the baseline TF-IDF model) via learning rate scheduling and dropout tuning.
- Engineered text normalisation and feature extraction improvements to the NLP preprocessing pipeline, automating the weekly reporting workflow and eliminating manual data entry.
- Deployed a real-time Streamlit dashboard backed by SQLite, automating sentiment aggregation for 10+ mentors across 5+ courses and reducing manual report generation time from hours to minutes.

Technical Skills

AI & GenAI: LangGraph, CrewAI, LangChain (LCEL), Agentic AI Systems, Retrieval-Augmented Generation, RAG Pipelines, LLMs, Prompt Engineering, NLP, BERT, Transformers, Semantic Chunking, Embedding Models, Vector Search (FAISS), OpenAI API, Mistral AI, DeepSeek V3, Gemini API, Tavily, NVIDIA NIM, LangSmith
ML Engineering: PyTorch, Scikit-learn, Hugging Face, Pandas, NumPy, Fine-tuning, Transfer Learning, Model Training, Hyperparameter Tuning, Feature Engineering, Data Preprocessing, Model Evaluation, Sentiment Analysis
Deployment & Infrastructure: Python, FastAPI, Docker, Streamlit, REST APIs, Redis, AWS (S3, EC2, Lambda), Git, CI/CD, Pydantic
Data & Storage: SQL, MongoDB Atlas, MySQL, PostgreSQL, SQLite, Vector Databases

Certifications

Building LLM Applications with Prompt Engineering | *NVIDIA Deep Learning Institute* — 2025

- Built LLM-powered applications using NVIDIA NIM (Llama 3.1) and LangChain with live API integration and Pydantic-structured output validation across a capstone agentic workflow.